assignment-01-code

August 13, 2026

1 Assignment I

1.1 Import External Libraries

[1]:
```
import numpy as np
```

1.2 5. Vector Inner Product Using Python

[2]:
```
v1 = np.array([[np.sqrt(3)], [1]])
v2 = np.array([[1], [1]])
inner_product = np.vdot(v1, v2)

print(f"Inner product: {inner_product:.3f}")
print("v1 Shape:", v1.shape)
print("v2 Shape:", v2.shape)
```

```
Inner product: 2.732
v1 Shape: (2, 1)
v2 Shape: (2, 1)
```

1.3 6. Inner Product with Complex Coefficient 1

[3]:
```
v1 = np.array([[1j], [1]])
v2 = np.array([[-1j], [1]])
inner_result = np.vdot(v1, v2)

print(f"Inner product: {inner_result:.3f}")
```

```
Inner product: 0.000+0.000j
```

1.4 7. Inner Product with Complex Coefficient 2

[4]:
```
v1 = np.array([[1 + 2j], [1]])
v2 = np.array([[1 - 1j], [1]])
inner_result = np.vdot(v1, v2)

print(f"Inner product: {inner_result:.3f}")
```

```
Inner product: 0.000-3.000j
```